Limits of Work Extraction under Local Thermal Operations: A Numerical Study of Ergotropic Gaps and Correlation Dynamics

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We present a comprehensive numerical investigation into the ergotropic dynamics of bipartite quantum systems under Local Thermal Operations (LTO). We simulate a diverse set of system-bath setups, including qubit-qubit, qubit-harmonic, and harmonic-harmonic system-bath pairs, to determine if local energy-conserving unitaries and independent thermal reservoirs can yield an ergotropy improvement for the initial state. By optimizing over the unitary via random sampling and the multi-restart Nelder-Mead algorithm, we probe the interplay between quantum correlations (Negativity, Discord) and the resulting ergotropic advantage. Our results provide robust numerical evidence for a general monotonicity principle: ergotropy is strictly non-increasing under LTO in non-degenerate regimes. Crucially, we find that even in the presence of local subsystem degeneracies and higher dimensions for the system-bath configurations we have considered, the LTO framework does not provide an ergotropic advantage.

I. PROBLEM STATEMENT

The fundamental challenge in quantum thermodynamics is to delineate the operational boundaries of quantum resources, such as coherence and correlations, in work extraction protocols. While *Global Thermal Operations* (TO) allow for the collective manipulation of a system and its environment [9], *Local Thermal Operations* (LTO) impose a strict locality constraint: each subsystem interacts only with its own independent thermal reservoir. This constraint raises a critical question: can correlations, ordinarily considered a global resource, be converted into extractable work through purely local thermal operations?

A. Ergotropy

In this project, we address the possibility of **ergotropy improvement** under LTO. The ergotropy $\mathcal{R}$ of a quantum state ρ with Hamiltonian H is defined as the maximum work extractable under a unitary evolution U:

$$\mathcal{R}(\rho, H) := \mathrm{Tr}(\rho H) - \min_U \mathrm{Tr}(U\rho U^\dagger H) \tag{1}$$

where the minimum is taken over all unitary operators U acting on the system's Hilbert space. Equivalently, it can be expressed in terms of the passive state ρ_p:

$$\mathcal{R}(\rho, H) = \mathrm{Tr}(\rho H) - \mathrm{Tr}(\rho_p H) \tag{2}$$

where $\rho_p = \sum_i r_i \left|\epsilon_i\right\rangle \left\langle\epsilon_i\right|$ is a passive state, obtained by rearranging the eigenvalues of ρ in descending order and pairing them with the ascendingly ordered eigenvalues of H [4]. The ergotropy is classified as **global** ($\mathcal{R}_G$) if the system is treated as a collective unit for unitary optimization, and **local** ($\mathcal{R}_L$) if the extractable work is constrained to local unitary operations acting independently on each subsystem.

B. The LTO Framework

The **Local Thermal Operation (LTO)** paradigm on a bipartite system ρ_{AB} is defined via a three-stage "dilate-interact-trace" sequence. Consider two independent thermal reservoirs, R_A and R_B, at inverse temperatures β_A and β_B. Each reservoir interacts locally with its respective subsystem (S_A or S_B) such that the local energy, defined by the Hamiltonian $H_i = H_{S_i} \otimes \mathbb{I}_{R_i} + \mathbb{I}_{S_i} \otimes H_{R_i}$ for $i \in \{A, B\}$, is individually conserved [8]. Assuming the two composite local systems remain non-interacting with one another, the LTO map Λ_{LTO} is constructed as follows:

1. **Dilation:** The initial system state ρ_{AB} is coupled to the baths, which are initially in their respective Gibbs states: $\gamma_{R_i} = e^{-\beta_i H_{R_i}}/Z_i$. The total initial state is $\rho_{\mathrm{tot}} = \rho_{AB} \otimes \gamma_{R_A} \otimes \gamma_{R_B}$.

2. **Local Interaction:** The system and baths undergo a joint transformation via local, energy-conserving unitaries $U = U_A \otimes U_B$. These unitaries must satisfy the commutation relation $[U_i, H_{S_i} + H_{R_i}] = 0$, ensuring that energy is only exchanged between a subsystem and its corresponding local bath.

3. **Reduction:** The bath degrees of freedom are traced out, yielding the final bipartite system state:

$$\sigma_{AB} = \mathrm{Tr}_{R_A,R_B}\left[(U_A \otimes U_B)\rho_{\mathrm{tot}}(U_A^\dagger \otimes U_B^\dagger)\right]. \tag{3}$$

C. Numerical Investigations

Our core inquiry is to determine whether, under this framework, the final ergotropy of the system $\mathcal{R}(\sigma_{AB})$ exceeds the initial ergotropy $\mathcal{R}(\rho_{AB})$. To rigorously test this, our investigation is structured around four primary

objectives:

1. Scaling with Quantum Resources: We investigate the scaling of the ergotropic change, $\Delta\mathcal{R}_G$ (global) and $\Delta\mathcal{R}_L$ (local), against measures of correlations like *Mutual information*, *Negativity* (entanglement measure for a two-qubit system) and *Quantum Discord*. Our goal is to determine if these correlations can be converted into an ergotropic advantage or if the LTO constraint fundamentally prevents their use.

2. Dynamics of the Ergotropic Gap: We analyze the evolution of the ergotropic gap, defined as $\delta\mathcal{R} = \mathcal{R}_G - \mathcal{R}_L$. Since this gap is a known detector of entanglement

3. System Architectures and Degeneracy: To ensure the robustness of our results, we test the LTO framework across three distinct system-bath configurations: (i) qubit–qubit, (ii) qubit–harmonic, and (iii) harmonic–harmonic. Additionally, we explicitly compare degenerate vs. non-degenerate Hamiltonians in the qubit–qubit case to determine if local degeneracies are sufficient to enable an ergotropy increase.

4. Numerical Verification of Monotonicity: Given the non-differentiable nature of the ergotropy function, we utilize an optimization strategy to find the global maximum of $\Delta\mathcal{R}$. By proving that $\max(\Delta\mathcal{R}) \leq 0$ across a vast ensemble of states and local unitaries, we provide numerical grounding for the conjecture that ergotropy is a monotone under LTOs.

II. NUMERICAL METHODOLOGY

To investigate whether ergotropy can increase under Local Thermal Operations (LTO), we developed a numerical framework to simulate system-bath interactions. Our approach is divided into distinct modules that handle state generation, local thermal operations, unitary optimization, and ergotropy calculations.

A. Quantum State Sampling (`state_gen.py`)

The reliability of our results depends on testing a wide variety of initial states. We developed a library to generate several classes of bipartite states [6]:

1. **Product and Separable States:** Used as a baseline to study systems with zero entanglement or purely classical correlations. Product states are constructed via the Kronecker product of two random local density matrices, $\rho_A \otimes \rho_B$. Separable states are generated as convex combinations, $\rho = \sum p_i \rho_{A,i} \otimes \rho_{B,i}$, where p_i is a normalized random weight.

2. **Pure Entangled States:** Generated to test if high initial entanglement provides a susceptibility for ergotropy improvement. These are generated as Haar-random vectors $|\psi\rangle$ in the composite Hilbert space, filtered by a positivity requirement on the Negativity $N(\rho)$. We also use Schmidt-decomposed states, $|\psi\rangle = \sqrt{\alpha}\,|00\rangle + \sqrt{1-\alpha}\,|11\rangle$, to systematically vary entanglement.

3. **Mixed Entangled States:** These can be constructed by mixing multiple pure entangled states with random weights or by performing a partial trace over an ancilla space. This ensures that we probe states with varying levels of both entanglement and purity.

4. **Werner States:** A specific family of states defined as $\rho_W = p\,|\Phi^+\rangle\langle\Phi^+| + \frac{1-p}{d^2}\mathbb{I}$, where $|\Phi^+\rangle$ is a Bell state. These are used to analyze the transition from entangled to separable regimes by varying the parameter p.

5. **Higher-Dimensional Harmonic States:** For harmonic oscillator architectures, we generate Hilbert-Schmidt uniform mixed states. This is achieved by sampling a Haar-random pure state on a d^2-dimensional space and performing a partial trace over a d-dimensional environment to obtain a reduced state in the truncated system space. These allow us to test whether higher dimensions permit a breach of monotonicity.

B. Thermodynamic Observables (`energy.py` & `lin_alg.py`)

The extraction of ergotropy is computed numerically by determining the energy of the *passive state*, ρ_p. A state is passive if its density matrix is diagonal in the energy eigenbasis and its eigenvalues are non-increasing with respect to energy.

1. *Spectral Reordering and Passive Energy*

In the `energy.py` module, we calculate the passive energy using the spectral ordering theorem. For a system with Hamiltonian H, the minimal energy achievable via unitary transformation is:

$$E_p = \sum_{j=1}^{d} \lambda_j^{\downarrow} \epsilon_j^{\uparrow} \tag{4}$$

where:

- $\lambda_j^{\downarrow}$ denotes the eigenvalues of the density matrix ρ sorted in **descending** order ($\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$).
- $\epsilon_j^{\uparrow}$ denotes the eigenvalues of the Hamiltonian H sorted in **ascending** order ($\epsilon_1 \leq \epsilon_2 \leq \cdots \leq \epsilon_d$).

Numerically, this is implemented using `numpy.linalg.eigvalsh` [1] to obtain the spectrum, followed by a stable sort. This reordering ensures that the largest populations are paired with the lowest energy levels, representing the state from which no further work can be extracted.

2. Global vs. Local Measures

The distinction between global and local ergotropy depends on the dimensions of the spectral sort:

1. **Global Ergotropy ($\mathcal{R}_G$):** The sort is performed over the full $d_A d_B$-dimensional Hilbert space using the total Hamiltonian $H_{tot} = H_A \otimes \mathbb{I} + \mathbb{I} \otimes H_B$.

2. **Local Ergotropy ($\mathcal{R}_L$):** We first compute the reduced density matrices $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and $\rho_B = \mathrm{Tr}_A(\rho_{AB})$ via `einsum` contractions. The passive energy is then calculated independently for each subsystem using their local Hamiltonians H_A and H_B.

3. Correlation Quantifiers

To track the relationship between work extraction and quantum resources, `lin_alg.py` includes routines for the following measures:

- **Mutual Information (I):** Quantifies total correlations (both classical and quantum) between subsystems A and B, calculated as $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, where $S(\cdot)$ is the von Neumann entropy.

- **Negativity ($\mathcal{N}$):** A rigorous measure of entanglement in the two-qubit scenario, calculated via the Peres-Horodecki criterion. It is defined as the sum of the absolute values of the negative eigenvalues of the partial transpose ρ^{T_B}.

- **Classical Correlations (C):** Defined as the maximum information obtainable about subsystem A by performing a local projective measurement on subsystem B. Numerically, this is implemented by maximizing the expression:

$$C(A:B) = S(\rho_A) - \min_{\{\Pi_i^B\}} \sum_i p_i S(\rho_{A|i}) \tag{5}$$

  where $\{\Pi_i^B\}$ is a set of rank-1 projectors on B. The minimization of the conditional entropy is performed over the Bloch sphere coordinates (θ, ϕ) using the L-BFGS-B algorithm with multiple random restarts to ensure global convergence.

- **Quantum Discord (Q):** Quantifies non-classical correlations beyond entanglement. It is calculated as the difference between the total mutual information and the optimized classical correlations:

$$Q(A:B) = I(A:B) - C(A:B) \tag{6}$$

  Since discord requires the optimization of C, it serves as a sensitive indicator of how quantumness is distributed or dissipated during the LTO process.

C. Unitary Parameterization (`unitary.py`)

The LTO framework requires that the local unitaries U_A and U_B preserve the total energy of their respective system-bath pairs. In the energy eigenbasis, this implies that U_i must be a block-diagonal matrix, where each block corresponds to a degenerate energy subspace of $H_i = H_{S_i} + H_{R_i}$.

1. Subspace Identification and Haar-Random Sampling

The `deg_unitary` function identifies these blocks by scanning the diagonal entries of the total Hamiltonian for indices that share the same energy within a tolerance of 10^{-10}. For a degenerate subspace of dimension k, a $k \times k$ sub-matrix is constructed. To populate these blocks for general numerical sampling, we utilize Haar-random unitaries generated via QR decomposition (`random_unitary`).

2. Mapping Unitaries for Optimization

To transition from random sampling to active optimization, we map the unitary blocks to a real parameter vector. Each $k \times k$ block is parameterized via the exponential map $U_k(\vec{\theta}) = \exp\Big(i\mathcal{H}(\vec{\theta})\Big)$, where $\mathcal{H}$ is a Hermitian matrix. A vector $\vec{\theta} \in \mathbb{R}^{k^2}$ is used to fill $\mathcal{H}$ such that:

- The first k entries of $\vec{\theta}$ define the real diagonal elements.

- The remaining $k(k-1)$ entries are used in pairs to define the real and imaginary parts of the upper triangle.

- Conjugate symmetry ($\mathcal{H}_{ji} = \mathcal{H}_{ij}^*$) is enforced to satisfy the Hermitian condition.

This mapping allows every set of real parameters $\vec{\theta}$ to automatically generate a valid energy-conserving unitary, which is verified in our code by asserting that $[U_i, H_i]$ and $(U_i U_i^\dagger - \mathbb{I})$ are below 10^{-10}.

D. The LTO Simulation Engine (`local_to.py`)

The numerical implementation of the local thermal operation is centered around the `LTO_step` function.

1. System-Bath Coupling and Index Reordering

The simulation begins by constructing a joint Hilbert space of the system and the reservoirs. For subsystems A and B and reservoirs R_A and R_B, the code performs a dilation: $\rho_{\text{ext}} = \rho_{AB} \otimes \gamma_{R_A} \otimes \gamma_{R_B}$, where γ_{R_i} are the Gibbs states of the harmonic oscillator baths. To apply the local unitaries U_A and U_B acting on the $S_i R_i$ subspaces, the joint state is reshaped and transposed from (S_A, S_B, R_A, R_B) to (S_A, R_A, S_B, R_B). This allows the interaction to be computed as a Kronecker product, $U_{tot} = U_A \otimes U_B$.

2. Reduction via Optimized Contractions

The final system state σ_{AB} is recovered by tracing out the bath degrees of freedom. We utilize optimized `einsum` contractions to perform the partial trace:

$$\sigma_{ac,eg} = \sum_{b,d} \rho'_{abcd,efgh} \delta_{bf} \delta_{dh} \tag{7}$$

where indices $\{b, f\}$ and $\{d, h\}$ represent the reservoirs R_A and R_B. This method bypasses general partial trace functions, ensuring faster execution and higher numerical stability.

3. Numerical Stability and Validation

The `sanity_check_gibbs` routine verifies that the LTO implementation respects thermodynamic laws. It ensures that a product Gibbs state $(\gamma_A \otimes \gamma_B)$ at the bath temperature remains a fixed point of the map. The engine validates the following conditions within a 10^{-10} tolerance:

- **Fixed Point Preservation:** Max error between input and output state is below threshold.
- **Bath Invariance:** Local bath states are unchanged post-interaction.
- **Trace Conservation:** The final density matrix σ_{AB} maintains a trace of 1.0.
- **Free Energy Monotonicity:** Verification that local free energies F_i do not increase.

E. Optimization Strategy (`optimize.py`)

To determine the maximum possible ergotropic gain, we implement a multi-staged numerical search for the optimal unitary in the LTO step. Since ergotropy is defined via spectral sorting, the resulting optimization landscape is non-differentiable at points of eigenvalue crossings, necessitating derivative-free methods.

1. Monte Carlo Global Search

We first perform a large-scale stochastic search (`search_over_states`). The algorithm generates an ensemble of N_s initial states and, for each, applies N_u Haar-random energy-conserving unitaries. This double-loop approach establishes the statistical distribution of ergotropy changes and identifies state configurations that may be susceptible to improvement.

2. Nelder-Mead Refinement

Promising states are subjected to a local optimization using the **Nelder-Mead simplex algorithm**(`nelder_mead_search`) [7]. We utilize the `minimize` routine from the `scipy.optimize` package [2] with the `method='Nelder-Mead'` option.

Numerical Loop: For a fixed state, the optimizer updates the flat parameter vector $\vec{\theta}$. In each step, the code reconstructs the unitaries U_A and U_B, performs the LTO simulation, and evaluates the change in ergotropy.

Global Convergence: To prevent the simplex from converging to local minima, we implement a `n_restarts` wrapper. Each restart initializes the search from a new random $\vec{\theta}_0$ scaled by 0.1 to begin near the identity operation.

We collect the results from all restarts and sampled states, maintaining the maximum achieved values for global ergotropy change ($\Delta\mathcal{R}_G$), local ergotropy change ($\Delta\mathcal{R}_L$), and the ergotropic gap ($\delta\mathcal{R}$). By verifying that the highest found value across all setups- qubit and harmonic, does not exceed a threshold of 10^{-8}, we provide a definitive numerical verification of the monotonicity of ergotropy under LTO for our given system-bath scenario.

F. Extensions to Harmonic Oscillator Setup

The numerical investigation was extended to higher-dimensional harmonic oscillator (HO) systems. In the HO regime, standard qubit-based entanglement measures such as Negativity are not applicable, and the optimization required for Quantum Discord becomes numerically intractable. Consequently, we rely on mutual information as the primary detector for total(classical and quantum) correlation changes. This regime introduces significant computational overhead and necessitates a shift in the metrics used to characterize state correlations.

1. Truncated Hilbert Space and Computational Complexity

To simulate harmonic systems, we utilize a truncated Hilbert space of dimension d. Unlike the qubit case, the complexity of unitary parameterization and density matrix operations scales significantly. As shown in our logs, a single Nelder-Mead restart in this regime can require approximately 30 minutes of wall-clock time. Despite this intensive search, the results remained consistent with the qubit case, yielding no ergotropic improvement ($\Delta\mathcal{R} \leq 0$).

2. Parameter Sweeps and Heatmaps

To ensure no specific resonance or temperature gradient could provoke a breach of monotonicity, we performed extensive parameter sweeps. We generated heatmaps for $\Delta\mathcal{R}_G$ and $\Delta\mathcal{R}_L$ across a grid of frequencies (ω_1, ω_2) and inverse temperatures (β_1, β_2). As illustrated in the Results section, these sweeps demonstrate that while the magnitude of ergotropy loss varies with the energy scales of the oscillators, the change remains strictly non-positive across the entire phase space.

G. Summary of the Numerical Workflow

The interaction between the modules is summarized in the following algorithmic sequence, designed to provide both statistical breadth and local optimization precision:

1. **Initialization:** Set local Hamiltonians (ω_1, ω_2), bath temperatures (β_A, β_B), and harmonic dimensions d.

2. **Global Resource Mapping:** For fixed system parameters, we generate a batch of N states from a specific class (e.g., Pure Entangled). We apply fixed or optimized LTO unitaries to this ensemble to observe how the changes in ergotropy ($\Delta\mathcal{R}$) and the ergotropic gap ($\delta\mathcal{R}$) scale with initial quantum resources.

3. **State-Specific Unitary Sweeps:** To analyze the sensitivity of ergotropy to correlation dynamics, we fix a single state (typically Haar-random) and apply a large ensemble of energy-conserving unitaries. This allows us to track the variation of $\Delta\mathcal{R}$ against the change in observables like Mutual Information and Negativity.

4. **Optimization and Monotonicity Testing:** Here, for each sampled configuration, we perform a rigorous search for ergotropic advantage using two methods, as aforementioned, Monte Carlo Search and Nelder-Mead Refinement.

III. RESULTS

We evaluate the performance of Local Thermal Operations (LTO) across four distinct system-bath configurations. In each scenario, we utilized either exhaustive random sampling or Nelder-Mead optimization to find a potential ergotropy increase. Our results consistently show that ergotropy change remains non-positive ($\Delta\mathcal{R} \leq 0$), supporting the conjecture that ergotropy is a monotone under the LTO framework. The analytical derivation proving $\Delta\mathcal{R} \leq 0$ for the non-degenerate case is provided in Appendix A.

A. Non-Degenerate Qubit-Qubit Setup

The first test case consists of a bipartite system where both the subsystems and their respective local reservoirs are qubits. We utilize non-degenerate local Hamiltonians with distinct frequencies, typically set around $\omega_1 = 1.0$ and $\omega_2 = 2.0$, with bath temperatures $\beta_A = 1.0$ and $\beta_B = 2.0$. To ensure the results are not state-dependent, we performed a sweep over a diverse ensemble of 500 initial states, including product, separable, Werner, and mixed entangled states. The following Fig. illustrates the change in global and local ergotropy $(\Delta\mathcal{R}_G, \Delta\mathcal{R}_L)$ plotted [3] against the initial state Negativity $\mathcal{N}$ and the "general" analytical bounds derived in Appendix B.

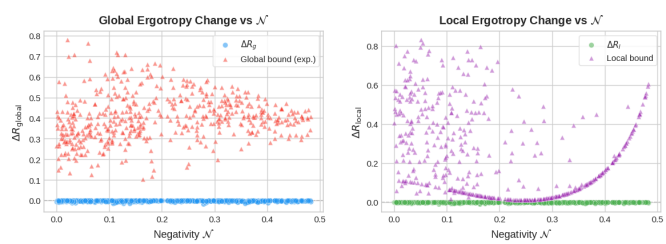

FIG. 1: **Ergotropy Change under LTO vs. Initial Negativity** in Non-Degenerate Qubit-Qubit Setup.

Global vs. Local Change: While the magnitude of ergotropy loss varies with the initial correlations, every sampled state sits strictly on or below the zero-change baseline.

Comparison with Analytical Bounds: The numerical data is compared against the theoretical bound $\Delta\mathcal{R} \leq 0$. Even for states with maximum initial Negativity ($\mathcal{N} = 0.5$), the local unitaries optimized via random sampling failed to provoke any ergotropic gain. With the best LTO unitary being Identity.

Since the results for the non-degenerate case are qualitatively identical across all state classes, we present only the aggregate random sampling data here to establish the baseline. Specific dynamics for separable and pure

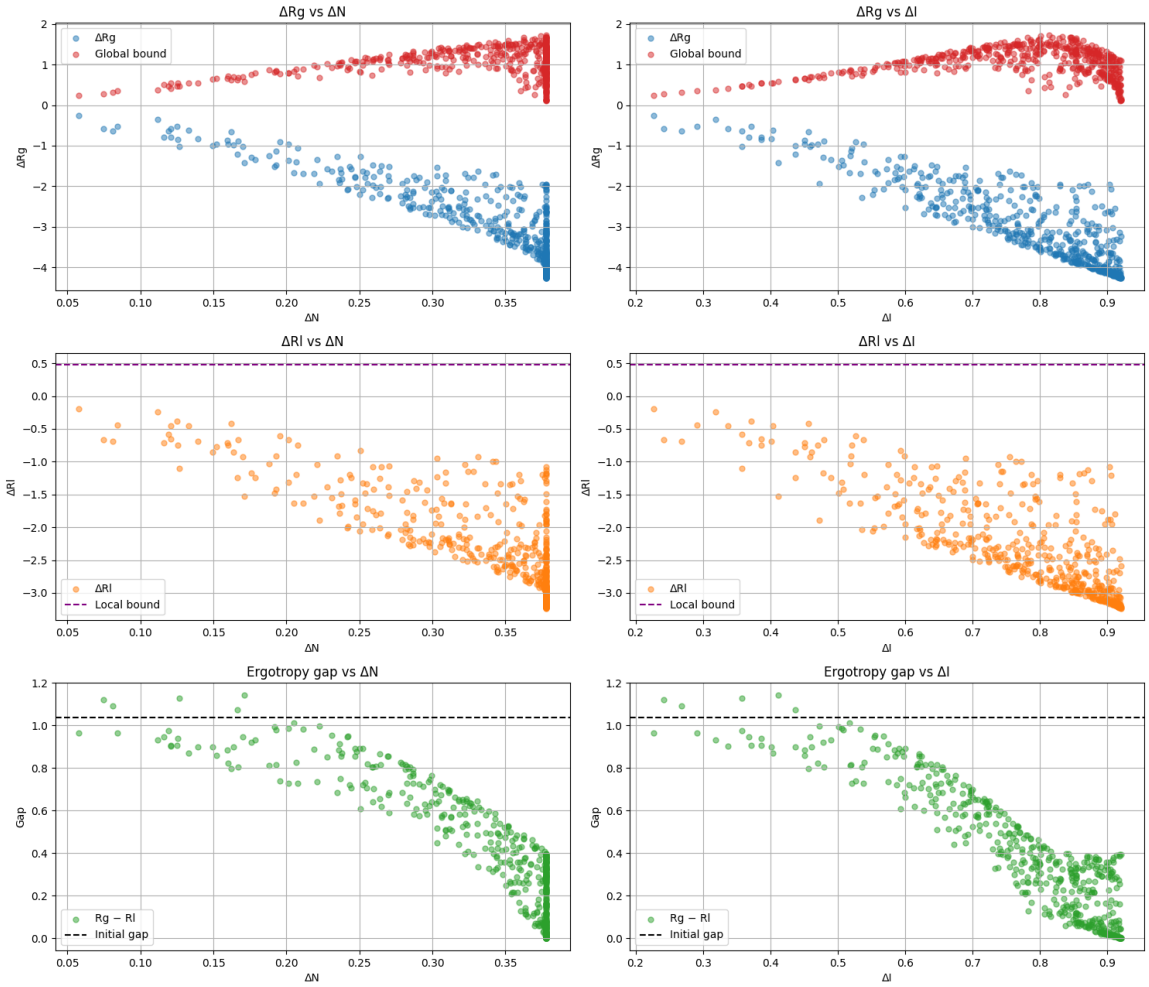

FIG. 2: **Ergotropy and Correlation Dynamics for a Fixed Pure Entangled State:** The three rows track the evolution of global ergotropy change $\Delta\mathcal{R}_G$, local ergotropy change $\Delta\mathcal{R}_L$, and ergotropic gap, against Negativity $\mathcal{N}$ and Mutual Information I, respectively. The analytical bounds are also drawn.

entangled states are detailed in the following sections to highlight their role in correlation-work trade-offs.

B. Degenerate Qubit-Qubit Setup

We investigate the ergotropy dynamics in the presence of local degeneracies between subsystem and bath hamiltonians, where the system and bath frequencies are matched ($\omega_{S_1} = \omega_{B_1} = 1.0$ and $\omega_{S_2} = \omega_{B_2} = 2.0$). This setup permits non-trivial energy-conserving transformations within the degenerate subspaces. Despite the increased operational freedom, numerical searches across all state classes yielded no instances of ergotropic improvement ($\Delta\mathcal{R} \leq 0$). While product and mixed states follow the same monotonic behavior, we focus our presentation on the results for **pure entangled states**. These states exhibit the most interesting dynamics between work extraction and quantum resources. Optimization results and additional plots for other state classes (separable, Werner, etc.) are omitted as they qualitatively mirror the "no-go" results of the pure entangled case. We perform two types of numerical sweeps:

1. **Unitary Ensemble Sweep (Fixed State):** We fix a single pure entangled state and apply a large ensemble of energy-conserving unitaries. This reveals the full range of accessible ergotropy changes and how they relate to the dissipation of entanglement and other correlation measures, as shown in Fig. 2 and 3.

2. **State Ensemble Sweep:** As shown in Fig. 4, we sweep over a large batch of pure entangled states, applying a fixed unitary or best-found unitary from our random sampling and Nelder-Mead routines.

C. Qubit System - Harmonic Bath Setup

In this scenario, we couple the qubit subsystems to truncated harmonic baths ($d = 10$), respectively. This

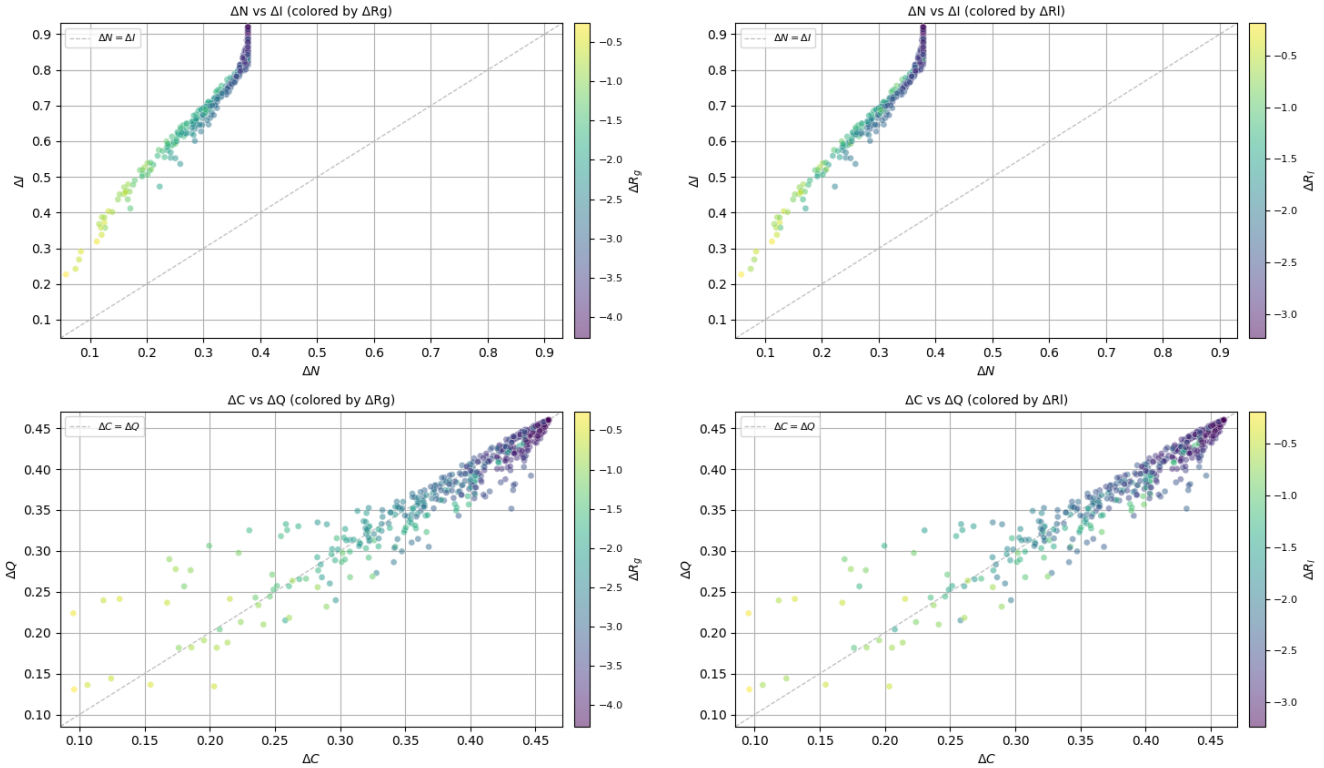

FIG. 3: **Resource Dissipation Trade-offs under LTO:** Phase-space plots mapping the simultaneous change in quantum and classical resources during the unitary sweep. (Top) The change in Mutual Information ΔI vs. Negativity $\Delta\mathcal{N}$, colored by ergotropy change. (Bottom) Comparison of Classical Correlation change ΔC vs. Quantum Discord change ΔQ, colored by ergotropy change.

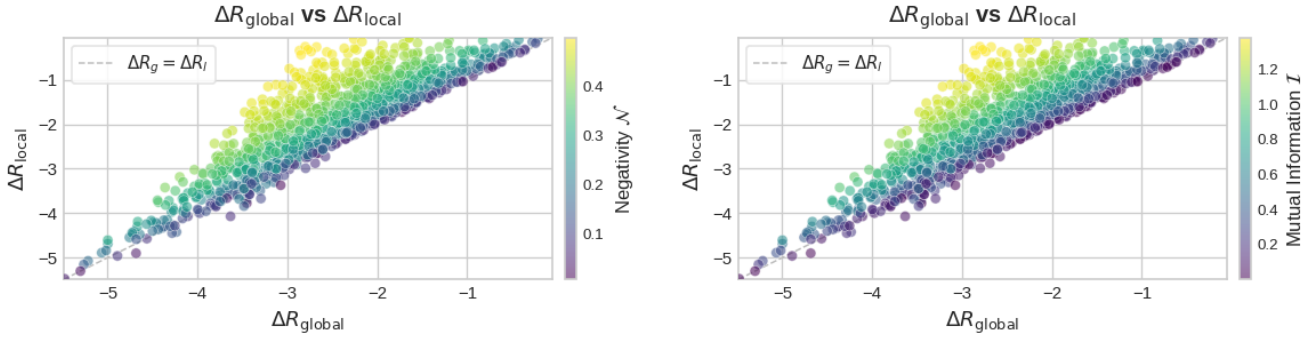

FIG. 4: **Global vs. Local Ergotropy Change across Correlation metrics:** Aggregate data for 500 pure entangled states, where each state is transformed by its fixed LTO map. The dashed line represents the $\Delta\mathcal{R}_G = \Delta\mathcal{R}_L$ identity with a point being in the lower triangle, implying an increase in the ergotropic gap. The data is colored by initial Negativity (Left) and Mutual Information (Right).

configuration was used to investigate whether the increased density of states in the reservoir could facilitate ergotropic advantage under LTO. We specifically probed till the limit of vanishingly small frequencies ($\omega_i, \omega_{R_i} \to 10^{-4}$) and temperatures ($\beta_A = 1.0, \beta_B = 0.9$) to identify potential breaches of monotonicity.

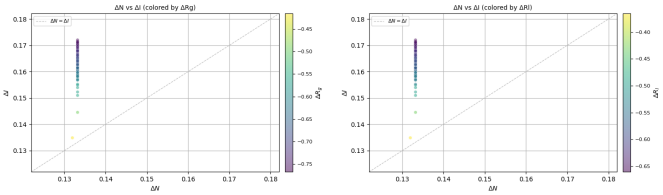

FIG. 5: ΔI **vs** $\Delta\mathcal{N}$**, colored by ergotropy change** in Qubit-Harmonic Setup.

As illustrated in Fig. 5, the results show a rapid decay of both entanglement ($\mathcal{N}$) and mutual information (I) during the LTO step. The high dimensionality of the harmonic bath acts as a powerful sink for correlations, ensuring that any local unitary transformation remains strictly contractive for ergotropy. To confirm that these results were not artifacts of poor sampling, we performed local refinement on the most promising states. A representative log of the Nelder-Mead optimization is provided below:

Optimization Output

```
Nelder-Mead:  100% |##########| 10/10 [05:03,
30s/restart]   restart 10:  Δglobal=-3e-06,
Δlocal=-3e-06, Δgap=0.000246
```

The optimization consistently converged to non-positive values. This confirms that the ergotropy monotone holds even when transitioning between differing physical configurations.

D. Harmonic-Harmonic Setup

The final architectural test involves a harmonic-harmonic configuration, where both the subsystems and their respective reservoirs are modeled as truncated harmonic oscillators. This setup allows for an increased density of states, which we utilized to test if specific frequency resonances or temperature gradients could facilitate an ergotropy increase.

Heatmap Parameter Sweeps: We performed extensive grid searches across the (ω, β) phase space to detect any potential breach of the ergotropy bound. The system was initialized with maximum dimensions $d_{sys} = 4$ and $d_{bath} = 10$, considering computational costs. We tracked the global and local ergotropy changes $(\Delta\mathcal{R}_G, \Delta\mathcal{R}_L)$ over $\omega \in [0.01, 0.2]$ to investigate the role of local energy level spacing and $\beta \in [0.1, 2.0]$ to assess the influence of bath coldness on work extraction.

Observation of Monotonicity: The results, presented as heatmaps in the following figure, show that the ergotropy change remains strictly non-positive across the entire sampled manifold. While the magnitude of ergotropy loss is highly sensitive to the frequency ω (as seen by the vertical "bands" of dissipation), the zero-increase bound is never violated.

As with the previous architectures, the harmonic-harmonic case qualitatively confirms that neither high-dimensional state density nor specific energy-scale matching can provoke a local ergotropic advantage. Consequently, we omit further individual state optimizations or correlation dynamics, as the aggregate phase-space data provides a comprehensive "no-go" result.

IV. INTERPRETATION OF RESULTS

The consistency of the numerical results across qubit, mixed, and harmonic architectures allows us to move beyond anecdotal simulation data toward a generalized physical interpretation of ergotropy dynamics under local thermal operations (LTO).

A. Monotonicity and the Role of Degeneracy

The most fundamental result of this study is the numerical verification of ergotropic monotonicity. The "No-Go" result in the non-degenerate case, as shown in Fig. 1, is also absolute and analytically grounded. As derived in Appendix C, the LTO constraint forces the interaction to reduce to a unital energy-dephasing channel. In this regime, the majorization of the output spectrum ($\lambda(\sigma) \prec \lambda(\rho)$) ensures that the passive energy can only increase, making the identity operation the trivial global optimum for work preservation. Our optimization results using both Monte Carlo sampling and Nelder-Mead refinement confirm that even when degeneracies are introduced (the qubit-degenerate and harmonic cases), the system cannot bypass this contractive behavior.

B. Resource Trade-offs and Correlation Dissipation

The relationship between work extraction and initial quantum resources is most clearly visualized in the plots for pure entangled states, particularly Figure 4.

Quadratic Clustering and Bounds: All data points reside strictly in the third quadrant, showing monotonicity under LTO, concentrated near the $\Delta\mathcal{R}_G = \Delta\mathcal{R}_L$ identity line. However, a significant distribution appears in the upper triangle ($|\Delta\mathcal{R}_L| < |\Delta\mathcal{R}_G|$), indicating an

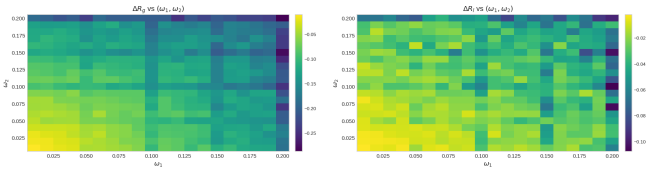

FIG. 6: **Frequency Dependence in Harmonic Systems.** Heatmaps of $\Delta\mathcal{R}_G$ and $\Delta\mathcal{R}_L$ vs. (ω_1, ω_2) showing that ergotropy loss is minimized near resonance but remains strictly ≤ 0.

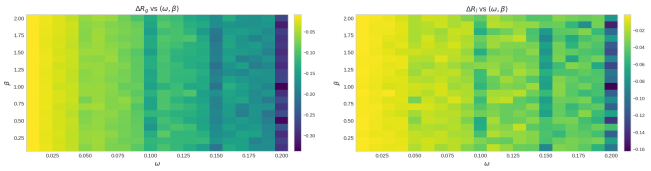

FIG. 7: **Temperature-Frequency Phase Space.** Sweeps across ω and β demonstrate the robustness of the ergotropy monotone against varying thermal gradients.

ergotropic gap decay. This demonstrates that the loss in global extractable work consistently exceeds the loss in local work, highlighting that LTO aggressively eliminates the non-local work potential stored in correlations.

Correlation Sensitivity and Dissipation Asymmetry: The color-coding by Negativity ($\mathcal{N}$) and Mutual Information (I) reveals a hierarchical dissipation structure that distinguishes local from global ergotropy loss. The high-correlation states (yellow regions) are positioned furthest from the origin in the $\Delta\mathcal{R}_G$ axis but remain relatively high in the $\Delta\mathcal{R}_L$ axis. This positioning implies that while entanglement decay is the more dominant occurrence under LTO, it primarily penalizes the global work potential. Because the decrease in classical correlations is less dominant than the damping of quantum correlations, the local ergotropy ($\Delta\mathcal{R}_L$) experiences a limited decrease compared to the global ergotropy ($\Delta\mathcal{R}_G$). This asymmetry facilitates the observation of points in the upper triangle where $\Delta\mathcal{R}_L > \Delta\mathcal{R}_G$, confirming that shared quantum resources are "locked" away from local extraction even as they are rapidly dissipated.

Irreversible Correlation Loss: The proximity of the points to the identity line at low correlation values implies that for weakly correlated states, the LTO map acts almost exclusively on local degrees of freedom. Conversely, for highly entangled states, the vertical spread (deviation from $\Delta\mathcal{R}_G = \Delta\mathcal{R}_L$) represents the "total loss" of shared resources. LTO treats both classical and quantum correlations as noise to be dissipated into the reservoirs rather than as fuel for local work.

C. Hierarchical Correlation Decay

To resolve the relationship between correlation dissipation and work potential, we examine the behavior of non-classical and classical markers during the LTO step. Figures 2 and 3 track these dynamics for a fixed pure entangled state subjected to an ensemble of energy-conserving unitaries. In Fig. 2, both $\Delta\mathcal{R}_G$ and $\Delta\mathcal{R}_L$ remain strictly below their respective theoretical bounds (red and purple markers). The plot shows that any interaction with the reservoir results in an irreversible decay. Crucially, the ergotropic gap (bottom row) dissipates toward zero for larger changes in negativity, indicating its role as a detector for quantum correlations.

Color-Mapped Dissipation: In Fig. 3, the coloring by $\Delta\mathcal{R}_G$ and $\Delta\mathcal{R}_L$ reveals a clear sequence in resource degradation. For low-dissipation unitaries (yellow points), the data follows a steep trajectory that initiates at a significant offset ($\Delta\mathcal{N} \approx 0.05, \Delta I \approx 0.23$). This vertical start indicates that a baseline of mutual information and entanglement is dissipated immediately upon engaging the LTO map. In this regime, the data remains nearly parallel to the ΔI axis, showing that entanglement damping occurs as an initial, mandatory cost of the interaction before classical correlation decay begins to scale.

Classical vs. Quantum Dominance: The bottom row of Fig. 3 reveals a critical distinction between the scale of correlation decay and its impact on ergotropy. While the change in Quantum Discord (ΔQ) is the more frequent and numerically dominant occurrence under LTO as evidenced by the high density of points in the upper triangle of the plot, the ergotropic change $\Delta\mathcal{R}$ shows a higher sensitivity to the dissipation of classical correlations (ΔC). The concentration of low-dissipation points (yellow/green) in the upper triangle near the origin, compared to their scarcity in the lower regions, suggests that the loss of classical mutual information drives ergotropy toward the zero-bound more aggressively than the damping of discord. This indicates that while LTO is a "quantum-destroying" process, the dissipation of classical shared resources is the primary physical driver behind the loss of global ergotropy.

D. Phase Space Analysis: Frequency and Temperature Sensitivity

The harmonic-harmonic heatmaps provide a comprehensive mapping of ergotropy dissipation across the frequency (ω) and inverse temperature (β) domains.

Frequency Resonances (ω): The vertical banding in Fig. 6 demonstrates that ergotropy loss is highly sensitive to the energy level spacing of the oscillators.

While dissipation is minimized at certain frequency ratios, the change $\Delta\mathcal{R}$ remains strictly non-positive across the entire frequency comb. This confirms that frequency tuning alone cannot bypass the LTO-induced work-extraction ceiling.

Invariance to Thermal Scaling (β): Crucially, for a fixed frequency, the ergotropy change $\Delta\mathcal{R}$ remains remarkably constant across the range of inverse temperatures $\beta \in [0.1, 2.0]$ as shown in Fig. 7. The heatmap reveals that increasing the bath temperature (lowering β) has a minimal effect on the magnitude of the ergotropic decay. This implies that the ergotropy monotone is not just a high-temperature limit result, but a structural property of the LTO map that persists regardless of the thermal energy scale.

V. CONCLUSION

Through an exhaustive numerical investigation across qubit, mixed, and harmonic architectures, we have characterized the dynamics of work extraction under the Local Thermal Operations (LTO) framework. Our findings provide a definitive numerical "No-Go" result: in the absence of global unitaries or non-thermal resources for our specific configurations. It shows that initial quantum correlations cannot be converted into a local or global ergotropic advantage. Even in the presence of subsystem degeneracies and high-dimensional reservoirs, the LTO constraint of local energy conservation acts as a fundamental filter that dissipates shared entanglement and mutual information without a corresponding ergotropic gain. The results consistently validate the theoretical bounds derived in Appendix A. Future research will focus on finding system–bath configurations that saturate the analytical bound.

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Appendix A: Source Code

The complete numerical suite is available at the following GitHub repository: Link

Appendix B: Thermodynamic Bounds on Ergotropy

In the framework of Local Thermal Operations (LTO), extractable work is bounded by the change in the local free energy. We define the total local free energy for a composite state ρ_{AB} as:

$$F_{\text{loc}}(\rho_{AB}) = F_A(\tilde{\rho}_A) + F_B(\tilde{\rho}_B) \tag{B1}$$

where $F_i(\tilde{\rho}_i) = E(\tilde{\rho}_i) - \beta_i^{-1} S(\tilde{\rho}_i)$. Under any LTO map Φ_{LTO}, the monotonicity of local free energy ($F_{\text{loc}}(\Phi(\rho)) \leq F_{\text{loc}}(\rho)$) and the minimization properties of Gibbs states lead to a fundamental bound on the change in local ergotropy $\Delta\mathcal{R}_L$:

$$\Delta\mathcal{R}_L \leq \sum_i \frac{1}{\beta_i} S(\tilde{\rho}_{i,P} \| \gamma_i) \tag{B2}$$

where $S(\cdot\|\cdot)$ is the relative entropy and $\tilde{\rho}_{i,P}$ is the local passive state [4]. Crucially, this bound contains no explicit correlation-dependent terms, suggesting that shared quantum resources cannot be converted into local ergotropy via LTO.

Similarly, the change in global ergotropy $\Delta\mathcal{R}_G$ is bounded by the change in local free energies and the stabilization of the passive state energy, yielding:

$$\Delta\mathcal{R}_G \leq \sum_i \frac{1}{\beta_i} S(\tilde{\rho}_i || \gamma_i) + [\text{Tr}(H\rho_p) - \text{Tr}(H\,|0\rangle\langle 0|)] \tag{B3}$$

where the second term represents the energetic distance between the initial passive state and the global ground state.

Appendix C: Monotonicity in Non-Degenerate Systems

For an n-partite system where the system-bath Hamiltonians are non-degenerate, the LTO constraint $[U_i, H_{S_i} + H_{R_i}] = 0$ restricts local unitaries to be diagonal in the energy eigenbasis. We prove that such operations reduce to energy-dephasing channels:

1. **Population Invariance:** The diagonal elements of the density matrix in the energy basis remain invariant: $\Phi_A(|i\rangle\langle i|) = |i\rangle\langle i|$.
2. **Coherence Damping:** Off-diagonal terms are damped by the factor $|\sum_t e^{i\phi_t}\gamma_t| \leq 1$.
3. **Majorization:** The resulting map ϕ is unital, implying the final state spectrum is majorized by the initial spectrum: $\lambda(\phi(\rho)) \prec \lambda(\rho)$.

By the Hardy-Littlewood-Pólya theorem, the passive energy satisfies $E_P(\phi(\rho)) \geq E_P(\rho)$. Consequently, ergotropy is strictly monotonic:

$$\Delta\mathcal{R}_G \leq 0 \quad \text{and} \quad \Delta\mathcal{R}_L \leq 0 \tag{C1}$$

This confirms that **subsystem degeneracy is a necessary condition** for any ergotropy increase.